

## Assignment 6

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### 1 How to prove that $3SAT \leq_P 3COLOR$

We prove that  $3SAT \leq_P 3COLOR$ , which implies  $3COLOR$  is NP-complete.

First,  $3COLOR \in NP$ . Given a coloring of the graph, we can check every edge and verify in polynomial time that its two endpoints receive different colors.

Now we reduce  $3SAT$  to  $3COLOR$ . We describe the construction using the following example:

$$\phi = (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z}).$$

The construction is built from three kinds of gadgets. We first describe each one in isolation and then explain how they are wired together.

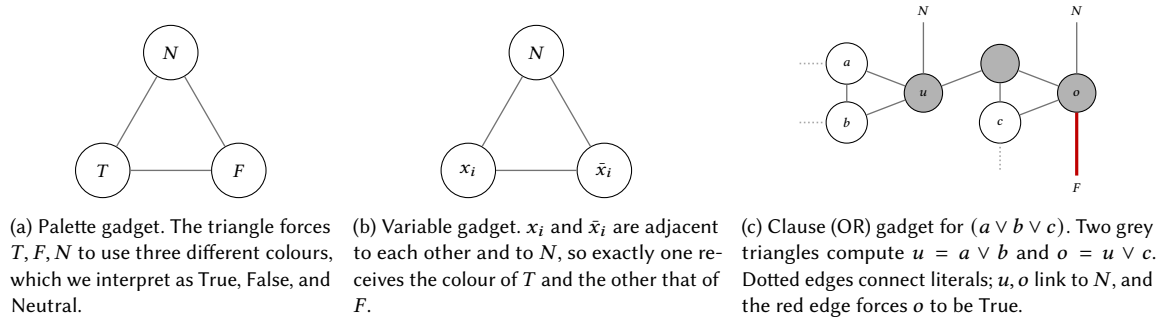


Fig. 1. The three elementary gadgets.

*Palette gadget (Figure 1a).* Three vertices  $T, F, N$  form a triangle. They must receive three distinct colours, which we call True, False, and Neutral. All other gadgets anchor to these three vertices.

*Variable gadget (Figure 1b).* For each Boolean variable  $x_i$  we create two vertices  $x_i$  and  $\bar{x}_i$ . Both are joined to  $N$  and to each other, forming a triangle with  $N$ . Because  $x_i$  and  $\bar{x}_i$  are adjacent to  $N$ , neither can take the neutral colour; because they are adjacent to each other, exactly one receives the colour of  $T$  (and is interpreted as True) while the other receives the colour of  $F$  (and is interpreted as False). Thus the gadget forces a consistent truth assignment.

*Clause gadget (Figure 1c).* For each clause  $(a \vee b \vee c)$  we chain two OR-gadgets:

- (1) The first grey triangle has inputs  $a, b$  and output  $u = a \vee b$ .
- (2) The second grey triangle has inputs  $u, c$  and output  $o = u \vee c$ .

The intermediate output  $u$  and the final output  $o$  are each connected to the palette vertex  $N$ , so both must be coloured either True or False. The final output  $o$  is also joined to the palette vertex  $F$  by a red edge. Since  $o$  is adjacent to  $F$ , it cannot itself be False; therefore  $o$  must be True. The internal structure of the OR-gadget guarantees that  $o$  can be True exactly when at least one of  $a, b, c$  is True. Hence the entire clause gadget is 3-colourable iff the clause is satisfied.

*Wiring the gadgets together.* Given a 3SAT formula  $\phi$ , the complete graph  $G_\phi$  is assembled as follows:

- (1) **Add the palette.** Insert the three palette vertices  $T, F, N$  and the edges among them.
- (2) **Add variable gadgets.** For each variable  $x_i$  in  $\phi$ , create vertices  $x_i$  and  $\bar{x}_i$ , add the edges  $(x_i, \bar{x}_i)$ ,  $(x_i, N)$ ,  $(\bar{x}_i, N)$ .
- (3) **Add clause gadgets.** For each clause  $C_j = (a_j \vee b_j \vee c_j)$  in  $\phi$ , instantiate a chain of two OR-gadgets with inputs  $a_j, b_j, c_j$ , intermediate output  $u_j$ , and final output  $o_j$  as described above. Connect  $u_j$  to  $N$ , connect  $o_j$  to  $N$ , and connect  $o_j$  to  $F$ .
- (4) **Connect literals to clauses.** Each literal vertex ( $x_i$  or  $\bar{x}_i$ ) is linked by a dotted edge to every OR-gadget input that corresponds to an occurrence of that literal in a clause. (These dotted edges are the only connections between the variable layer and the clause layer.)

The construction uses a constant number of vertices and edges per variable and per clause, so  $G_\phi$  has size polynomial in  $|\phi|$  and can be built in polynomial time.

Figure 2 shows the complete graph for the example formula  $\phi = (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z})$ .

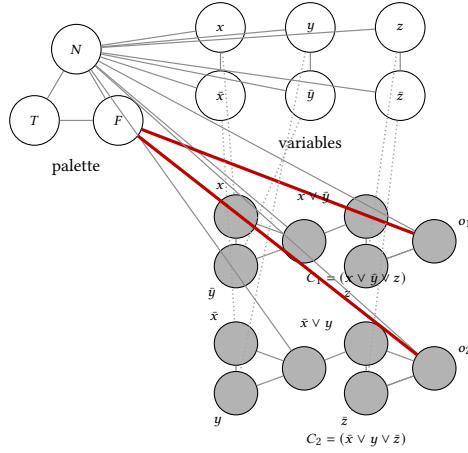


Fig. 2. Reduction from 3SAT to 3COLOR for the formula  $\phi = (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z})$ . Palette vertices  $T, F, N$  enforce the three colors; variable gadgets force a consistent truth assignment; clause gadgets (gray vertices) propagate the OR of literals and force the final output to be True via the red edges.

*Correctness.* If  $\phi$  is satisfiable, colour each literal vertex according to its truth value under a satisfying assignment (colour of  $T$  for True, colour of  $F$  for False). Because every clause has at least one true literal, each OR-gadget can propagate True to its final output  $o_j$ , which is compatible with the red edge to  $F$  and the edge to  $N$ . Hence  $G_\phi$  admits a proper 3-colouring.

Conversely, suppose  $G_\phi$  has a proper 3-colouring. The variable gadgets force each pair  $(x_i, \bar{x}_i)$  to receive opposite non-neutral colours, so we read off a Boolean assignment by interpreting the colour of  $T$  as True. If a clause  $C_j$  were unsatisfied under this assignment, all three of its literal inputs would be coloured False. The OR-gadgets would then force the final output  $o_j$  to be False as well. But  $o_j$  is adjacent to  $F$  (red edge), so  $o_j$  cannot take the colour of  $F$ —a contradiction. Therefore every clause is satisfied and  $\phi$  is satisfiable.

The reduction is polynomial-time because each variable and each clause contributes only  $O(1)$  vertices and edges. Thus  $3\text{SAT} \leq_P 3\text{COLOR}$ . Since 3SAT is NP-complete and  $3\text{COLOR} \in \text{NP}$ , we conclude that 3COLOR is NP-complete.